

6(a)	<i>Vibrio <u>cholerae</u></i> ;	1
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Question	Answer	Marks
6(b)	any three from: 1 cholera already, endemic / present, in area ; 2 contaminated water / water contains pathogen ; R polluted water 3 pathogen transmitted by, faecal–oral route / AW ; 4 ref. to changed living conditions ; e.g. no piped, treated, drinking water / disruption to water treatment plants no safe disposal of sewage / disruption to sewage treatment plants 5 ref. to inability to practise good hygiene ; e.g. washing (utensils) in contaminated water not able to wash hands after defaecation latrines not covered 6 AVP ; e.g. ref. to flies transferring pathogen from, uncovered latrines / (contaminated) faeces	3
6(c)(i)	any three from: 1 mRNA unable to bind ; 2 ribosome subunits unable to come together ; 3 tRNA unable to, enter / bind to, ribosomes ; 4 mRNA codon–tRNA anticodon binding prevented ; 5 translation, prevented / does not begin or polypeptide synthesis prevented / polypeptides not synthesised ; A protein for polypeptide 6 enzymes required for growth not produced / example of role of protein in growth e.g. for membrane formation ;	3
6(c)(ii)	cell wall ;	1

Question	Answer	Marks
6(d)	any four from: <i>vaccination programme to max 3:</i> 1 vaccines stimulate (active) immunity to cholera / AW ; 2 idea that if infected with, pathogen / <i>V. cholerae</i>, people do not become ill with cholera or the immune system destroys pathogen before it can, cause harm / spread ; 3 ref. to <u>herd immunity</u> ; A described e.g. enough people effectively immunised to prevent spread idea that unvaccinated people surrounded by immunised people 4 AVP ; e.g. idea that reduces reservoir of pathogen in population <i>treatment to max 3:</i> 5 (oral / intravenous) rehydration therapy, decreases recovery time / cures more quickly ; 6 (more likely that) contaminated faeces disposed of safely / decreased risk of spread via contaminated faeces 7 more able to practise good hygiene ; 8 time spent in medical centre acts as quarantine / infected people not in general population ; 9 (so) decreases, density / proportion, of infected people to reduce risk of spread ; 10 AVP ; e.g. medical centres able to monitor antibiotic use and provide most effective medication medical centres, accessible to many parts of the country / widespread idea that medical centres mean people treated sooner decreasing time when people are infectious injecting antibodies	4